Parallel lines and transversals

ANSWER KEY

Core Concepts: Parallel lines and transversals

1 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 82° ✓

b. 92°

c. 74°

d. 127°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 45^\circ) = 82^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm ✓

c. 21 cm

d. 28 cm

Multiplying the original length by the scale factor gives $7 \times 2 = 14$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 11

c. 9 ✓

d. 12

Substituting $x = 1$ into $y = 3x + 6$ gives $y = 3(1) + 6 = 9$.

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 75° ✓

b. 85°

c. 67°

d. 127°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 52^\circ) = 75^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 2$ into $y = 3x + 6$ gives $y = 3(2) + 6 = 12$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 68° ✓

b. 78°

c. 60°

d. 127°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 59^\circ) = 68^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm ✓

c. 35 cm

d. 112 cm

Multiplying the original length by the scale factor gives $7 \times 4 = 28$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 12

b. 17

c. 15 ✓

d. 18

Substituting $x = 3$ into $y = 3x + 6$ gives $y = 3(3) + 6 = 15$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 61° ✓

b. 71°

c. 53°

d. 127°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 66^\circ) = 61^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 15

b. 20

c. 18 ✓

d. 21

Substituting $x = 4$ into $y = 3x + 6$ gives $y = 3(4) + 6 = 18$.

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